

MATPLOTLIB

1. What is Matplotlib and why is it used?

Matplotlib is a Python plotting library used to create graphs and visualizations such as line charts, bar charts, and scatter plots. It is used to analyze data visually, identify patterns, and present results clearly.

2. Difference between line plot, bar plot, and scatter plot

Line plot – Shows trends or changes in data over time

Bar plot – Compares values between different categories

Scatter plot – Shows the relationship between two variables

3. What are figure and axes in Matplotlib?

Figure – The overall window or canvas that holds the plot

Axes – The area where the data is plotted with x-axis and y-axis

4. How do labels, titles, and legends improve data visualization?

Labels describe the x-axis and y-axis

Title explains what the graph represents

Legend identifies different data sets in the plot

They make graphs clear, readable, and easy to understand.

5. Why is data visualization important before building AI models?

Data visualization helps to understand data distribution, detect outliers, identify patterns, and find errors. This improves data quality and leads to better and more accurate AI models.